

Worksheet 3 Solutions
MTH 203 (Multivariate Calculus)
Date: 01-09-2026

- (1) Suppose $f(x, y) = 5x^2 + y^2$. Using the definition of partial derivatives, compute f_x and f_y . Also, evaluate $f_x(2, 3)$ and $f_y(2, 3)$.

Solution: Using the definition of the partial derivative with respect to the variable x ,

$$\begin{aligned} f_x(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(5(x_0 + h)^2 + y_0^2) - (5x_0^2 + y_0^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5((x_0 + h)^2 - x_0^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5(2x_0h + h^2)}{h} \\ &= \lim_{h \rightarrow 0} (10x_0 + 5h) \\ &= 10x_0. \end{aligned}$$

Thus,

$$\boxed{f_x(2, 3) = 20.}$$

By parallel calculations, simply interchanging the roles of x_0 and y_0 , one obtains

$$f_y(x_0, y_0) = 2y_0,$$

which implies

$$\boxed{f_y(2, 3) = 2 \cdot 3 = 6.}$$

- (2) Show that $f_x(0, 0) = 0$, but $f_y(0, 0)$ does not exist, where

$$f(x, y) = \begin{cases} (x^2 + y) \sin\left(\frac{1}{x^2 + y^2}\right), & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Provide a detailed analysis.

Solution: Using the definition we have

$$\begin{aligned} f_x(0, 0) &= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h^2)}{h} \\ &= \lim_{h \rightarrow 0} h \sin(1/h^2). \end{aligned}$$

Since, $|h \sin(1/h^2)| \leq |h|$ we have,

$$f_x(0, 0) = 0.$$

Similarly:

$$\frac{f(0, k) - f(0, 0)}{k} = \frac{k \sin(1/k^2)}{k} = \sin(1/k^2).$$

The expression $\sin(1/k^2)$ has no limit as $k \rightarrow 0$. Therefore,

$$\boxed{f_y(0, 0) \text{ does not exist.}}$$

(3) Find $\frac{\partial z}{\partial x}$ if the equation

$$yz - \ln z = x + y$$

defines z as a function of two independent variables x and y and the partial derivative exists.

Solution: Given

$$yz - \ln z = x + y,$$

where $z = z(x, y)$.

Differentiating partially with respect to x , treating y as a constant, we get

$$y \frac{\partial z}{\partial x} - \frac{1}{z} \frac{\partial z}{\partial x} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial x} = 1 + 0.$$

Therefore,

$$\left(y - \frac{1}{z}\right) \frac{\partial z}{\partial x} = 1.$$

Hence,

$$\boxed{\frac{\partial z}{\partial x} = \frac{1}{y - \frac{1}{z}} = \frac{z}{yz - 1}}.$$

(4) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a given linear transformation. Compute $\frac{\partial T}{\partial x}$, $\frac{\partial T}{\partial y}$ and $\frac{\partial T}{\partial z}$ at the point (x_0, y_0, z_0) .

Solution: Using the definition of partial derivatives and the linearity of T , we have

$$\begin{aligned} \frac{\partial T}{\partial x}(x_0, y_0, z_0) &= \lim_{h \rightarrow 0} \frac{T(x_0 + h, y_0, z_0) - T(x_0, y_0, z_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{T(h, 0, 0)}{h} \\ &= \lim_{h \rightarrow 0} T(1, 0, 0) \\ &= T(1, 0, 0). \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{\partial T}{\partial y}(x_0, y_0, z_0) &= \lim_{h \rightarrow 0} \frac{T(x_0, y_0 + h, z_0) - T(x_0, y_0, z_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{T(0, h, 0)}{h} \\ &= T(0, 1, 0). \end{aligned}$$

Finally,

$$\begin{aligned}\frac{\partial T}{\partial z}(x_0, y_0, z_0) &= \lim_{h \rightarrow 0} \frac{T(x_0, y_0, z_0 + h) - T(x_0, y_0, z_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{T(0, 0, h)}{h} \\ &= T(0, 0, 1).\end{aligned}$$

Hence,

$\frac{\partial T}{\partial x}(x_0, y_0, z_0) = T(1, 0, 0), \quad \frac{\partial T}{\partial y}(x_0, y_0, z_0) = T(0, 1, 0), \quad \frac{\partial T}{\partial z}(x_0, y_0, z_0) = T(0, 0, 1).$
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(5) Verify that

$$T(x, t) = e^{-t} \cos\left(\frac{x}{c}\right)$$

satisfies the heat equation

$$\frac{\partial T}{\partial t} = c^2 \frac{\partial^2 T}{\partial x^2}.$$

Solution. Differentiate with respect to t :

$$\frac{\partial T}{\partial t} = -e^{-t} \cos\left(\frac{x}{c}\right).$$

Differentiate twice with respect to x :

$$\frac{\partial^2 T}{\partial x^2} = e^{-t} \frac{d^2}{dx^2} \cos\left(\frac{x}{c}\right) = e^{-t} \left(-\frac{1}{c^2}\right) \cos\left(\frac{x}{c}\right).$$

Thus,

$$c^2 \frac{\partial^2 T}{\partial x^2} = -e^{-t} \cos\left(\frac{x}{c}\right) = \frac{\partial T}{\partial t},$$

(6) By direct calculation, show that

$$f_{xyz} = f_{yzx} = f_{zyx}$$

for the function

$$f(x, y, z) = xyz + x^2 y^3 z^4.$$

Solution: Given

$$f(x, y, z) = xyz + x^2 y^3 z^4.$$

First, differentiate f with respect to x :

$$f_x = yz + 2xy^3 z^4.$$

Next, differentiate f_x with respect to y :

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = z + 6xy^2 z^4.$$

Now differentiate f_{xy} with respect to z :

$$f_{xyz} = \frac{\partial}{\partial z}(f_{xy}) = 1 + 24xy^2 z^3.$$

1. Computation of f_{yzx} :

First differentiate with respect to y :

$$f_y = xz + 3x^2y^2z^4.$$

Next, differentiate with respect to z :

$$\begin{aligned} f_{yz} &= \frac{\partial}{\partial z} (xz + 3x^2y^2z^4) \\ &= x + 12x^2y^2z^3. \end{aligned}$$

Finally, differentiate with respect to x :

$$\begin{aligned} f_{yzx} &= \frac{\partial}{\partial x} (x + 12x^2y^2z^3) \\ &= 1 + 24xy^2z^3. \end{aligned}$$

Thus,

$$\boxed{f_{yzx} = 1 + 24xy^2z^3}.$$

2. Computation of f_{zyx} :

First differentiate with respect to z :

$$f_z = xy + 4x^2y^3z^3.$$

Next, differentiate with respect to y :

$$\begin{aligned} f_{zy} &= \frac{\partial}{\partial y} (xy + 4x^2y^3z^3) \\ &= x + 12x^2y^2z^3. \end{aligned}$$

Finally, differentiate with respect to x :

$$\begin{aligned} f_{zyx} &= \frac{\partial}{\partial x} (x + 12x^2y^2z^3) \\ &= 1 + 24xy^2z^3. \end{aligned}$$

Hence,

$$\boxed{f_{zyx} = 1 + 24xy^2z^3}.$$

Therefore,

$$\boxed{f_{xyz} = f_{yzx} = f_{zyx} = 1 + 24xy^2z^3}.$$

- (7) Define differentiability of a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ in a rectangular domain D . Further, show that if

$$f(x, y)$$

is differentiable at (x_0, y_0) , then it is also continuous at (x_0, y_0) .

Solution:

Definition. A function $z = f(x, y)$ is **differentiable** at (x_0, y_0) if $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist and

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

satisfies an equation of the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y,$$

in which each of $\epsilon_1, \epsilon_2 \rightarrow 0$ as both $\Delta x, \Delta y \rightarrow 0$.

We call f **differentiable** if it is differentiable at every point in its domain.

Differentiability implies continuity.

Since

$$\Delta f = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1(\Delta x) + \varepsilon_2(\Delta y),$$

consider the limit as

$$(\Delta x, \Delta y) \rightarrow (0, 0).$$

Because $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ are fixed finite numbers,

$$f_x(x_0, y_0)\Delta x \rightarrow 0, \quad f_y(x_0, y_0)\Delta y \rightarrow 0.$$

Also, since

$$\varepsilon_1(\Delta x) \rightarrow 0, \quad \varepsilon_2(\Delta y) \rightarrow 0,$$

Consequently,

$$\Delta f \rightarrow 0.$$

That is,

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \rightarrow 0.$$

Hence,

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} f(x_0 + \Delta x, y_0 + \Delta y) = f(x_0, y_0).$$

Therefore, f is continuous at (x_0, y_0) .

Differentiability of f at (x_0, y_0) implies continuity at (x_0, y_0) .

(8) Given

$$z = u(x, y)e^{ax+by}$$

and

$$\frac{\partial^2 u}{\partial x \partial y} = 0,$$

find the values of the constants a and b such that

$$\frac{\partial^2 z}{\partial x \partial y} - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} + z = 0.$$

Solution: Given

$$z = u(x, y)e^{ax+by}, \quad \frac{\partial^2 u}{\partial x \partial y} = 0.$$

First, differentiate z with respect to x :

$$z_x = e^{ax+by}(u_x + au).$$

Similarly,

$$z_y = e^{ax+by}(u_y + bu).$$

Now differentiate z_y with respect to x :

$$\begin{aligned} z_{yx} &= \frac{\partial}{\partial x} [e^{ax+by}(u_y + bu)] \\ &= e^{ax+by}(bu_x + u_{xy} + abu + au_y). \end{aligned}$$

Since $u_{xy} = 0$,

$$z_{yx} = e^{ax+by}(bu_x + au_y + abu).$$

Substituting z_{xy} , z_x , z_y and z into

$$z_{xy} - z_x - z_y + z = 0,$$

we get

$$\begin{aligned} 0 &= e^{ax+by} [bu_x + au_y + abu - u_x - au - u_y - bu + u] \\ &= e^{ax+by} [(b-1)u_x + (a-1)u_y + (ab-a-b+1)u]. \end{aligned}$$

Since

$$ab - a - b + 1 = (a-1)(b-1),$$

we have

$$(b-1)u_x + (a-1)u_y + (a-1)(b-1)u = 0.$$

For this relation to hold for the general class of functions satisfying $u_{xy} = 0$, we require

$$b-1=0 \quad \text{and} \quad a-1=0.$$

Hence,

$$\boxed{a=1, \quad b=1}.$$